

Chapter 166

Resolve FX Temporal

DaVinci Resolve comes with several Resolve FX that focus on the application of an effect over time.

Contents

Motion Blur (Studio Version Only).....	3600
Motion Trails (Studio Version Only).....	3600
General	3600
Advanced Options.....	3600
Move Trail.....	3601
Smear (Studio Version Only).....	3601
Stop Motion	3602

Motion Blur (Studio Version Only)

This effect replicates the Motion Blur panel in the Color page's Motion Effects palette. The Resolve FX version allows you to use these tools across the other pages in the program. Motion Blur settings use optical flow-based motion estimation to add artificial motion blur to clips that have none. This can be useful in cases where a program was shot using a fast shutter speed, and you later decide that the resulting video has too much strobing. By analyzing the motion within a clip, the Motion Blur settings can selectively apply blurring to the image based on the speed and direction of each moving element within the scene.

Three parameters let you set how much motion blur to add, and at what quality:

- **Motion Est. Type:** A setting of Better provides more accurate pixel mapping at the expense of being more processor intensive. Faster provides a more approximate result but is less processor intensive.
- **Motion Range:** Determines what speed of motion to consider when defining regions being blurred.
- **Motion Blur:** Raise this parameter to add more motion blur to the image, lower it to add less. The range is 0–100, where 0 applies no motion blur, and 100 applies maximum motion blur.
- **Blur Direction:** Allows you to choose which way the blur extends from the current frame. The options are: Both Directions, From Previous Frame, and Towards Next Frame.
- **Granularity:** Adjusts how much detail is added to the blur.

Motion Trails (Studio Version Only)

The Motion Trails effect copies the image to create ghost-like trails on moving images. This effect can be used to simulate clips shot with long shutter speeds and analog video feedback effects.

General

The first two sliders control the number and strength of the copied frames.

- **Trail Length:** Determines the number of copies that are used to create the trails.
- **Dropoff:** Sets the fade applied to each copied frame. The Dropoff value is compounded. For instance, using a value of .5 applies 50% opacity to the first copy, 25% to the second, 12.5% to the third, and so on.

Advanced Options

The advanced options are used to control how the overlapping copied frames are blended.

- **Composite Gamma:** The Composite Gamma menu provides four options for controlling the brightness of the overlapping frames.
 - Timeline:** Uses the Timeline Color Space setting in the Project Settings to control the overlapping brightness. This setting, by default, is Rec. 709, Gamma 2.4.
 - Rec. 709:** Uses a Rec. 709 color space with Gamma 2.2 to control the overlapping brightness.
 - Linear:** Uses Linear Gamma, often producing much stronger highlights.
 - Custom:** Provides a custom gamma slider which defaults to Gamma of 2.4. Setting this to a value of 1.0 is the same as setting the menu to Linear.

- **Input Alpha:** The choices in the Input Alpha menu determine how the Alpha channel is used for blending the frames. All options assume there is an Alpha channel present on the clip that has the Motion Trail applied.

Ignore: This is the default option and will cause the alpha channel to be ignored. Disabling the Use Alpha checkbox when Ignore is selected will cause the Alpha to act as a stencil effect for the trails.

Use in Compositing: This option uses the Alpha channel for the trails and compositing over the lower video track, assuming the Use Alpha checkbox is enabled.

Move Trail

This group of controls determines the offset, direction, size, and rotation of the copies. Adjustment amounts are applied as offsets from one copy to the next. For instance, if rotation is set to 5 degrees, the first trail copy is rotated 5 degrees, the second trail copy is rotated 10 degrees, the third copy is rotated 15 degrees, and so on.

- **Pan:** Adjusts the offset of the trail copies from the original image.
- **Pan Angle:** Changes the angle at which the trails are offset from the original image. The values are shown in degrees.
- **Zoom:** Adjust a scale adjustment, successively, to each copy.
- **Rotate:** Applies an angle of rotation, successively, to each copy.
- **Reuse Current Frame:** When this checkbox is disabled (the default), each copy uses the frame after the copy before it. For instance, the first copy is the current frame -1, and the second copy is the current frame -2. When this checkbox is enabled, all copies use the same frame as the current frame in the clip.
- **Border Type:** This menu determines how edges of the frame are handled when the copies are scaled smaller than the timeline resolution.

Black: The area outside the image is set to black.

Soften: This control blends the edges of the rectangular frame borders to give the images a more organic appearance.

Replicate: Duplicates the outermost pixels along the edge of the image. The pixels are stretched out from each side to reach the timeline resolution boundary.

Reflect: The image is flipped and flopped to create a mirrored image that extends to the timeline resolution frame boundary.

Wrap-Around: Duplicates the image to create a video wall effect, used to fill in the space to the timeline resolution frame boundary.

Smear (Studio Version Only)

The Smear effect simulates motion blur in a clip by blending a user-definable number of frames.

- **Frames Either Side:** Determines the number of frames that are blended on either side of the current frame. For instance, entering a value of 2 uses two frames before and two frames after the current frame to create the effect.
- **Luma Threshold:** This control sets how bright a pixel must be to contribute to the smear effect. A lower value causes only darker pixels to be smeared. A higher value includes brighter pixels in the effect.
- **Chroma Threshold:** This control sets how saturated a pixel must be to contribute to the smear effect. A lower value causes only low saturated pixels to be smeared. A higher value includes more saturated pixels in the effect.

Stop Motion

This plugin is used to replicate the stuttering motion effect found in stop motion animations. Traditional stop motion animation exposes each frame of the shot individually, while the subject is manually moved in tiny increments between frames. An extremely labor intensive process, often individual frames were repeated in a shot to speed up the production which resulted in a staccato motion cadence.

Advanced Options

Set Reference Frame: Selects the current frame in the Viewer as the reference frame.

Reference Frame: The frame number of the reference frame used for the skip pattern. The frame number is calculated from the start of the entire original media clip (frame 0).

Sampling Variation: This slider increasingly produces more uneven motion in the clip.

Frame Repeat: Selects how many frames (from 1-10) to duplicate in the shot. A larger number makes a more staggered cadence.